

Sarmad Mehrdad



Summary

Sarmad Mehrdad received his Ph.D. in Electrical Engineering at New York University on January 2026, advised by Prof. Ludovic Righetti. His research focuses on inverse reinforcement learning and optimal control for human–robot interaction, with applications in human motion prediction and real-time task intent inference. His work has been published in *IEEE RA-L*, *IROS*, *Humanoids*, *EMBC*, and *Nature Scientific Reports*, and he is a co-inventor on a patented wearable IoT health-monitoring device.

Research Interests: Inverse Reinforcement Learning, Optimal Control, Human–Robot Interaction, Reinforcement and Imitation Learning, Probabilistic Modeling.

Education

- **Ph.D.** New York University, Electrical Engineering June 2020 – January 2026
- **M.Sc.** Clarkson University, Mechanical Engineering January 2017 – August 2019
- **B.Sc.** K. N. Toosi University of Technology, Mechanical Engineering August 2010 – August 2015

Research Experience

- **Real-Time Cost Learning**
Developing online IRL methods for dynamic adaptation of cost-function estimates during human corrective interactions (Work in progress).
- **Inverse Reinforcement Learning**
Developed an iterative Minimal-Observation IRL (MO-IRL) framework for cost-function estimation, eliminating the need for initial non-optimal local sampling and increasing computational speed by over 200% (Lead author on *IROS 2025*; *IEEE RA-L* manuscript in preparation).
- **Human Motion Learning and Prediction**
Modeled human reaching and lifting behaviors using MO-IRL with Crocoddyl/Pinocchio/CasADi and validated on motion-capture datasets (Author on *Humanoids 2025*; lead author on *arXiv 2025*; *IEEE RA-L* manuscript in preparation).
- **Previous Topics**
 - Probabilistic modeling and Learning from Demonstration (*arXiv 2023*).
 - AI for health deterioration prediction in respiratory patients (*Nature Sci. Rep. 2023*; *Frontiers 2021*).
 - Co-inventor, NYU BioTracker wearable for COVID-19 tracking (US20240081657A1; WO2022241311A1).
 - EMG/EEG/HD-EMG data collection for rehabilitation and fatigue assessment (*IEEE TBME 2022*; *EMBC 2023 and 2024*; *Physiol Rep 2024*; *bioRxiv 2021*).
 - 3D-printed and silicone-based soft actuators for wearable robotics (*IEEE RA-L 2021*; *arXiv 2023*).

Publications (After 2020)

- **Mehrdad, Sarmad**, Sabbah Maxime, Vincent Bonnet, and Ludovic Righetti. "Toward Global Intent Inference for Human Motion by Inverse Reinforcement Learning" *arXiv:2603.07797*, 2026.
- Sabbah, Maxime, Filip Bečanović, **Sarmad Mehrdad**, Ludovic Righetti, Bruno Watier, and Vincent Bonnet. "Minimal Observations Inverse Reinforcement Learning for Predicting Human Box-Lifting Motions." In *International Conference on Humanoid Robots (Humanoids 2025)*.
- **Mehrdad, Sarmad**, Sabbah Maxime, Vincent Bonnet, and Ludovic Righetti. "Learning Human Reaching Optimality Principles from Minimal Observation Inverse Reinforcement Learning" *arXiv:2510.00329*, 2025.
- **Mehrdad, Sarmad**, Avadesh Meduri, and Ludovic Righetti. "Cost Function Estimation Using Inverse Reinforcement Learning with Minimal Observations." In 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 6071-6078. IEEE, 2025.
- O'Keefe, Rory, Neha Mehta, Yair Shahrar, **Sarmad Mehrdad**, Anat Lubetzky, and S. Farokh Atashzar. "Measuring Changes in Cortical Processes during VR-aided Physical Exercise—an EEG-Based Approach." In 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 1-5. IEEE, 2024.
- Yang, Jinghui, Rory O'Keefe, Seyed Yahya Shirazi, **Sarmad Mehrdad**, S. Farokh Atashzar, and Smita Rao.

"Muscle activity and hypoalgesia in blood flow restricted versus unrestricted effort-matched resistance exercise in healthy adults." *Physiological Reports* 12, no. 14 (2024): e16037.

- Libby, Jacqueline, Aniket A. Somwanshi, Federico Stancati, Gayatri Tyagi, **Sarmad Mehrdad**, JohnRoss Rizzo, and S. Farokh Atashzar. "How Does the Inner Geometry of Soft Actuators Modulate the Dynamic and Hysteretic Response?." *arXiv:2308.04722*, 2023.
- **Mehrdad, Sarmad**, Farah E. Shamout, Yao Wang, and S. Farokh Atashzar. "Deep learning for deterioration prediction of COVID-19 patients based on time-series of three vital signs." *Scientific Reports* 13, no. 1 (2023): 9968.
- **Mehrdad, Sarmad**, and S. Farokh Atashzar. "FiMReSt: Finite Mixture of Multivariate Regulated Skew-t Kernels–A Flexible Probabilistic Model for Multi-Clustered Data with Asymmetrically-Scattered Non-Gaussian Kernels." *arXiv:2305.09071*, 2023.
- O’Keeffe, Rory, Vaibhavi Rathod, Seyed Yahya Shirazi, **Sarmad Mehrdad**, Alexis Edwards, Smita Rao, and S. Farokh Atashzar. "Linear versus Nonlinear Muscle Networks: A Case Study to Decode Hidden Synergistic Patterns During Dynamic Lower-limb Tasks." In *2023 11th International IEEE/EMBS Conference on Neural Engineering (NER)*, pp. 01-05. *IEEE*, 2023.
- O’Keeffe, Rory, Seyed Yahya Shirazi, **Sarmad Mehrdad**, Tyler Crosby, Aaron M. Johnson, and S. Farokh Atashzar. "Perilaryngeal-Cranial Functional Muscle Network Differentiates Vocal Tasks: A Multi-Channel sEMG Approach." *IEEE Transactions on Biomedical Engineering* (2022).
- A. Altalbani, H. Zhou, **S. Mehrdad**, F. Alambeigi, and S. F. Atashzar, "Design, Fabrication, and Validation of a New Family of 3D-Printable Structurally-Programmable Actuators for Soft Robotics," in *IEEE Robotics and Automation Letters*, doi: 10.1109/LRA.2021.3101860.
- **Mehrdad, Sarmad**, Yao Wang, and S. Farokh Atashzar. "Perspective: Wearable Internet of Medical Things for remote tracking of symptoms, prediction of health anomalies, implementation of preventative measures, and control of virus spread during the era of COVID-19." *Frontiers in Robotics and AI* 8 (2021): 610653.
- O’Keeffe, Rory, Seyed Yahya Shirazi, Jinghui Yang, **Sarmad Mehrdad**, Smita Rao, and S. Farokh Atashzar. "Non-parametric Functional Muscle Network as a Robust Biomarker of Fatigue." *bioRxiv* (2021).
- O’Keeffe, Rory, Seyed Yahya Shirazi, **Sarmad Mehrdad**, Tyler Crosby, Aaron M. Johnson, and S. Farokh Atashzar. "Characterization of cervical-cranial muscle network in correlation with vocal features." *bioRxiv* (2021).
- **Mehrdad, S.**, Liu, F., Pham, M.T., Lelevé, A. and Atashzar, S.F., 2021. "Review of Advanced Medical Telerobots." *Applied Sciences*, 11(1), p.209.

Patents

- **Co-inventor:** *Smart Wearable IoT Device for Health Tracking, Contact Tracing, and Prediction of Health Deterioration*. US 20240081657 A1 (USPTO, 2024); WO 2022241311 A1 (WIPO, 2022), with inventorship amended to include S. Mehrdad.

Teaching Experience

- **New York University**
 - **Teaching and Course Assistant:** Haptics and Telerobotics; Interactive Medical Robotics 2021–2023
 - **K-12 Outreach:** ARISE; SPARC 2022–2024
- **Clarkson University**
 - **Instructor:** Mechanical and Aeronautical Engineering Labs 2018
 - **Teaching Assistant:** Robotics I & II, Dynamical Systems, Aircraft Design II, MAE & ECE Labs 2017–2019

Awards & Services

- **Awards**
 - **Ernst Weber Fellowship**, Tandon School of Engineering, New York University 2022
 - **Tandon School of Engineering Fellowship**, New York University 2020 & 2021
 - **K-12 STEM Education Fellowship**, Tandon School of Engineering, New York University 2020 & 2021
- **Academic Service**
 - **Member**, IEEE Robotics and Automation Society
 - **Reviewer**, IEEE T-RO, RA-L, ICRA, IROS, ICORR, ICAS 2021–2024
 - **Reviewer**, Elsevier Robotics and Autonomous Systems 2021